

Sprint 2 — Estimation Appendix

CS 5388 / CS 4390 Software Project Management

Team: 1 **Project:** Damn Parking **Date:** October 5, 2026

1) Baseline and Estimate Boundary

This appendix consolidates the September 29, 2026 estimate and the October 5 scope clarification. The estimated slice is Live Parking Monitoring: the same six stories previously summarized as “Parking Platform.” The baseline is four people at 20 project hours per person per week. The budget sheet’s five-person, 30-hour schedule and 10–15-lot, 60-camera launch are not used for this estimate.

Story	Priority	Points
Find nearby parking lots	Must	6
Compare live availability	Must	8
Get directions to a selected lot	Could	3
View parking patterns and possible misuse	Must	5
Filter by distance, type, and cost	Should	3
View a lot’s available-space count	Must	7
Total	All six included	32

The slice uses a responsive website with parking lookup/filter controls, lot details, directions, and a manager dashboard. It stores lot information and timestamped occupancy history. External dependencies are mapping/routing data and an existing occupancy-data service.

Excluded: camera installation, new computer-vision development or training, payments, reservations, automated enforcement, and the future large camera deployment. User departure reporting and future-availability prediction from the broader starter list are not included. The estimate therefore does not cover every Sprint 1 charter success criterion. Human review of possible misuse is assumed; detailed detection requirements could change the estimate.

2) Estimation Approach

Function Points start from functional counts. Story Points start from team-assigned relative sizes and an assumed velocity. Both estimate the same slice and include the same integration allowance. Counts and productivity remain preliminary assumptions, not measured implementation performance.

3) Function Point Calculation

Category	Count	Weight	FP
External inputs	1	4	4
External outputs	3	5	15
External inquiries	1	4	4
Internal logical files	2	10	20
External interface files	1	7	7
Total			50

Count basis: occupancy updates are one input. Calculated availability, manager summaries, and presented directions are three outputs. Nearby-lot lookup and filtering are one inquiry. Lot information and occupancy history are two internal data groups; external map/routing information is one external data group. Map and list presentations are not counted twice. Directions are assumed to be presented within the platform; a simple external link would require a count review.

Assumed rates are 6, 9, and 12 hours/FP. The activity suggests the 6–12 range; 9 is the selected midpoint. Base effort is $50 \times 6 = 300$, $50 \times 9 = 450$, and $50 \times 12 = 600$ hours. Adding 40 hours gives 340 / 490 / 640 hours.

4) Story Point Calculation

Assume two-week sprints, four people, and 40 hours per person per sprint: $4 \times 40 = 160$ team hours/sprint.

Velocity is assumed at 8 / 10 / 16 points per sprint; no completed implementation sprints support these values yet.

Scenario	Velocity	Sprints	Base hours	With 40 h
Low effort	16	$32 \div 16 = 2$	320	360
Middle	10	$32 \div 10 = 3.2$	512	552
High effort	8	$32 \div 8 = 4$	640	680

Base hours = sprints $\times$ 160 team hours. Story points are not assigned a universal hours-per-point rate.

Capacity and assumed velocity apply together to this particular team; changing headcount does not preserve velocity automatically.

5) Integration and Team Size

Additional work	Team hours
Occupancy data, storage, and calculations	10
Interface, map, directions, and dashboard integration	12
End-to-end testing and integration fixes	8
Coordination and handoffs	8
Unallocated portion of stated allowance	2
Total assumed allowance	40

The source worksheet lists $10 + 12 + 8 + 8 = 38$ hours but applies a 40-hour allowance throughout its calculations. This appendix preserves that 40-hour baseline and identifies the remaining 2 hours as unallocated; it does not increase a source line item. Base estimates cover feature work and feature-level testing; the allowance covers cross-feature integration and coordination and is added once per method. Four people have $4 \times (4 - 1) \div 2 = 6$ communication paths. No larger-team timeline is copied, and no proportional productivity increase from additional staff is assumed.

6) Reconciliation and Final Estimate

Function Points give 340–640 hours with a 490-hour middle estimate. Story Points give 360–680 hours with a 552-hour middle estimate. Their middle-value ratio is $552 \div 490 = 1.13$. Agreement is a cross-check, not independent validation, because scope and the 40-hour allowance are shared.

Productivity is the strongest tested driver: the FP rate changes effort by 300 hours across its range, while assumed velocity changes effort by 320 hours. Better interface definitions, reviewed functional counts, and representative integration estimates could narrow the range. The starter's whole-release comparison of 2–4 months is broader and uncalibrated; it is not substituted for this slice estimate.

Combined range: 340–680 hours. With $O = 340$, $M = 552$, and $P = 680$, PERT expected effort = $(340 + 4 \times 552 + 680) \div 6 = 538$ hours. M uses the more cautious story-based middle scenario; this is a planning judgment.

At the activity example's assumed loaded rate of \$60/hour, labor cost is \$20,400–\$40,800, with an expected \$32,280. At 80 team hours/week, duration is 4.25–8.5 weeks, with expected duration 6.73 weeks. These are projected implementation effort and labor costs, not actual Sprint 2 expenditure.